Cala · Usability and Accessibility

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CALA · Usability and accessibility project 1 UX/UI PROJECT · VACATION RENTAL PLATFORM cove Usability evaluation, accessibility and inclusive design Test plan, prototype evaluation and improvement proposals Document Academic project Study object Cala prototype: explore, details, reservation and confirmation Date October 3, 2026 Accessibility goal WCAG 2.2, level AA Summary The project documents the inspection of the Cala prototype and designs an inclusive usability study. If They analyze the understanding of the journey, economic transparency and access barriers. It is proposed a protocol with eight participants, privacy measures and two emerging technologies. Scope of results: The findings come from the HTML, CSS and JavaScript provided. have not been conducted sessions with participants nor a browser execution check. They don't show up fabricated metrics or testimonials.

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1. Context, scope and objectives Cala represents a vacation rental application through four screens: explore accommodations, consult the details, review the reservation and receive confirmation. The file contains low and high versions fidelity. The only programmed behavior is the fidelity change using the setFi(mode) function. The Search, filters, cards, dates, and booking and payment actions are static representations. The four screens are displayed together; there is no functional navigation path.

Evaluation objectives

1. Check if you understand how to search for and select accommodation.

2. Verify that services, dates, guests and total price are identified.

3. Detect doubts about payment, conditions and confirmation.

4. Identify visual, auditory, motor and cognitive barriers.

5. Prioritize improvements and define how to verify their effectiveness.

Two stages of study Stage What allows you to evaluate Limit Current prototype Comprehension, identification of information and intention of use Pointing out a control is not equivalent to completing a reservation Interactive version future Task success, errors, time and use of assistive technologies Requires implementing navigation and test controls Project structure

2. Method and participants · 3. Tasks and indicators · 4. Development and analysis · 5. Findings and improvements · 6. Privacy · 7. Inclusive design · 8. Accessibility · 9. Emerging technologies · 10. Conclusions and annexes.

CALA · Usability and accessibility project 3 Visual presentation of the prototype The following screenshots, contributed by the user, represent the two versions of the Cala interface. Both organize the tour in four screens: explore, accommodation details, reservation and confirmation.

Figure 1. Low fidelity wireframes Representation of the structure, hierarchy and distribution of content using boxes, lines and markers of image. Source: screenshot provided by the user.

CALA · Usability and accessibility project 4 Visual presentation: interface design Figure 2. High fidelity interface Application of visual identity: colors, typography, cards, economic summary and actions planned in each screen. Source: screenshot provided by the user. The screenshots document the appearance of the interface; do not prove by themselves that the actions of search, reservation or payment are functional nor that sessions have been held with participants.

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2. Method, participants and sessions An individual moderated test with observation and thinking aloud is proposed, followed by short questions. If verbalizing interferes with assistive technology or causes fatigue, retrospective questions. The moderator will facilitate the tasks without indicating where the solution is located. The proposed sample includes eight adults with different ages, digital experience and support needs. It is a qualitative study: it is not intended to statistically represent the entire population.

population.

Code Planned profile P01-P02 Two people with different digital experience, without declared support needs P03 Person with low vision using magnification P04 Blind person using screen reader P05 Person with a motor disability who uses a keyboard or alternative input P06 Deaf or hearing loss person P07 Person with comprehension, attention or reading needs P08 Older person with little digital experience Profiles may overlap. You will ask what adaptations each person needs, without requiring diagnoses. You will be allowed to use your usual device and settings.

Phase Indicative duration Information, consent and settings 5 minutes Context of use 5 minutes Tasks 25 minutes Assessment and final interview 10 minutes A pilot session will be held first to adjust the script. Breaks and more time will be offered when be necessary. Duration will not be a condition of participation.

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3. Tasks and success indicators Understanding tasks for the current prototype Task Expected result T1. Identify pool and pet admission Recognize both services at Villa Cala Gat T2. Explain how you would proceed to book Identify the Reserve action T3. Interpret the room August 12-18, 2 guests and 6 nights T4. Explain the full cost €1,367: stay, cleaning and service T5. Interpret the confirmation Recognize accommodation, dates and total T6. Locate subsequent actions Identify itinerary and contact with the host The moderator will present the high fidelity view. These tasks assess understanding and intention to use. The Static actions will be recorded as non-executable, not as completed functional tasks.

Indicators and internal goals Indicator Registration Proposed objective Understanding Correct answer without help At least 7 out of 8 per task Full price Identifies total and charges 8 out of 8 before confirming Future Functional Success Complete without help At least 7 out of 8 Future reservation time Simulated search to confirmation Median less than 3 minutes Perceived ease Question from 1 to 7 after each task Median of at least 5 Critical barriers Blocks that prevent you from continuing None pending In the functional version you will try searching, filtering, opening an accommodation, modifying dates and guests, reviewing the cost and simulate the payment. Fictitious data will be used without charges. Goals are not normative standards; They will be reviewed after the pilot.

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4. Development and analysis of tests Preparation The prototype, script, adaptations and consent will be checked. It will be explained: “We are evaluating the design; You can stop or ask for a break whenever you want.” Notifications will be disabled personal and fictitious scenarios will be used.

Log during tasks The observer will record participant code, task, response, time when applicable, errors, help, barriers and relevant comments. You will classify the result as success without help, success with help, abandonment or not evaluable. The absence of functionality will be recorded as a limitation of the prototype.

Planned analysis Incidents will be grouped by task and probable cause. Each problem will include evidence, participants affected, impact, proposal and subsequent verification. Individual counts will be reported and medium; Percentages will not be generalized with such a small sample.

Severity Criterion Criticism Prevents you from completing an essential task or accessing your information High Causes major errors or requires help to continue Medium It produces doubts or effort, although it allows you to continue Low Affects comfort or presentation without impeding the task The results with assistive technologies will be analyzed according to the barriers found. A longer time does not will automatically be attributed to the person's capacity. Design problems, limitations will be separated of the prototype and conditions of the session.

Execution status Completed: reading and inspection of the code and review of economic consistency. Pending: sessions, task measurements, subjective assessment, browser testing and full audit. They don't exist yet satisfaction data or user testimonials.

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5. Findings and proposals for improvement Code finding Impact and improvement Priority Search and reservation controls without logic They prevent the flow from being evaluated. Implement navigation and dummy data.

Criticism Tabs div with onclick, without keyboard or control semantics They prevent standard keyboard access. Use native buttons with focus and accessible state.

High Price per night until checkout Difficult to compare the full cost. Show total for dates chosen from the list.

High Dates, guests and payment static They do not allow you to correct selections. Create labeled controls and validation.

High Fixed heights and overflow:hidden Risk of cuts when expanding. Allow growth or displacement and check visually.

High; check No meta viewport Risk of inappropriate mobile presentation. Add it and review adaptation.

Medium No visible cancellation policy Lack of information to decide. Show conditions before payment.

High Photographs replaced by degraded Limits confidence evaluation. Incorporate images representative and alternative.

Medium The breakdown is arithmetically consistent: €210 × 6 nights + €45 cleaning + €62 service = €1,367.

The confirmation keeps the same total. Priorities are inspection estimates. Its frequency and real impact will be contrasted with participants. Correcting controls is the first step to be able to measure effectiveness and efficiency.

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6. Data collection and privacy Only participant code, approximate digital experience, requested adaptations, task results and comments. An observation sheet and a short questionnaire will be used. The screen or audio recording will be optional and will have separate authorization; rejecting it will not prevent participate. Before the sessions, you will be informed about the person responsible, purpose, legal basis, recipients, conservation and exercise of rights. The GDPR requires minimization and transparency. The institution must specify these data before starting the study [3].

Measurement Application Pseudonymization Codes P01-P08; contacts and authorizations in a separate file Fictitious data Do not ask for personal accounts, card numbers or actual reservations Limited access Encrypted institutional folder, accessible only to the evaluation team Minimum recording Avoid notifications, personal desktop and information unrelated to the test Proposed conservation Recordings: 30 days; pseudonymized records: 90 days. Confirm before the study Presentation of results Aggregated data and de-identified quotes; review re-identification risks External services Do not send recordings to IA without evaluating treatment and informing previously Pseudonymization is not equivalent to anonymization. Medical diagnoses will be avoided: it will be enough to know the necessary adaptations. If it is essential to process health data, the conditions will be reviewed applicable specifics. The HTML loads fonts from Google Fonts. It is proposed to house them locally to reduce dependencies external. The inspection does not identify forms, analytics, or participant storage, but this does not replaces a review of network requests.

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7. Inclusive design Need Adaptations and checks visual Screen reader, magnification, contrast, headers and image alternatives. Verify reading the total and confirmation.

Auditory Written instructions and text communication. If videos are incorporated, provide subtitles and relevant alternatives.

motor Full keyboard or alternative input, comfortable objectives and actions that do not require precision or drag.

Cognitive Clear language, predictable steps, summary before payment, errors with a solution and the possibility of review without losing data. Adaptations will be agreed with each participant. Instructions will be available in writing and They will allow breaks. Imposing a single form of interaction will be avoided.

Physical accommodation accessibility The accessibility of the interface does not guarantee that of the accommodation. Cala should offer verified information on entrance without steps, elevator, widths and adapted bathroom. A generic “accessible” label does not Provides enough detail to decide.

Inclusive acceptance criteria Essential tasks must be able to be completed with assistive technologies. The relevant content must remain available when expanding. The economic information will have a clear textual formulation and the Confirmations will not depend only on color or an icon. After the improvements, the evaluation will be repeated with people who found barriers, when possible. If You will also verify that the changes do not generate difficulties in other forms of use.

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8. Accessibility: WCAG 2.2 AA objective Level AA is established as an objective because it includes relevant requirements for different forms of perception and interaction. Conformity requires satisfying all applicable criteria A and AA. The prototype cannot be declared compliant currently: there are barriers in the code and complete checks are missing [1].

Principle Evaluation and improvement Perceptible Darken secondary text. Review image alternatives, enlargement and redistribution. Criteria relevant: 1.1.1, 1.3.1, 1.4.3, 1.4.4 and 1.4.10.

Operable Replace tabs with accessible controls and verify keyboard, order and focus. Check size of objectives and non-hidden focus. Criteria: 2.1.1, 2.4.3, 2.4.7, 2.4.11 and 2.5.8.

Understandable The Spanish language is declared. Explain conditions, errors and consequences; allow to correct before payment. Criteria: 3.1.1, 3.3.1, 3.3.2 and 3.3.4.

Robust Use semantics, names, roles and accessible states. Announce confirmation whenever dynamic. Criteria: 4.1.2 and 4.1.3.

Contrast calculated from CSS colors Text/background Relationship Normal text #6E7B76 / #FFFFFF 4.41:1 insufficient #6E7B76 / #EFF2ED 3.91:1 insufficient #142B2E / #FFFFFF 14.84:1 Enough #5B3E06 / #E3A62F 4.57:1 Enough Criterion 1.4.3 requires 4.5:1 for normal text [2]. The calculations refer to these declared combinations; they do not test all rendered combinations or global conformance. review will be combined automatic, manual and supporting technologies.

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9. Proposed emerging technologies 9.1. Multimodal conversational assistant with AI It could make searching via text or voice easier and explain prices in plain language. I would help people with difficulties reading, data entry or understanding filters. It is presented as a future proposal, not as existing functionality. Design conditions: Maintain conventional controls, display selections before applying them, and allow them to be corrected. Accommodation accessibility data must come from verified information. The assistant must not invent them or confirm payments autonomously. Validation: compare comprehension and errors with and without assistance; check that people can Leave the conversation without losing your selections. Review the treatment of queries and avoid unnecessary conservation.

9.2. Speech recognition and automatic transcription It could allow you to search for accommodations without writing and transcribing audio messages. Voice input can help people with motor difficulties; transcription can facilitate access to messages for people who are deaf or have hearing loss. Design Conditions: Offer text as an alternative, ask for microphone permission, and allow editing and confirm what is recognized. Evaluate errors with different accents and forms of speech. Prefer processing local when feasible. Validation: recording recognition errors, correction effort and transcription understanding.

Economic operations should require explicit review.

Inclusive integration These technologies should expand the available options without becoming a mandatory path. Its usefulness It will be validated with users and will not replace the basic accessibility of the site.

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10. Conclusions and improvement plan Cala presents a comprehensible conceptual sequence and a coherent economic breakdown. The inspection identifies static controls, keyboard barriers, insufficient contrast in secondary text, and risks of presentation, along with opportunities to show the total and conditions further in advance. An inclusive protocol has been designed and the available inspection results have been documented. The actual sessions and the complete audit are pending. The report does not attribute participants results that have not yet been collected.

Order Action Check 1 Implement controls and simulated tour Search and reservation tasks can be executed 2 Fix semantics, keyboard and contrast Manual review and assistive technologies 3 Improve total price, conditions and edition Understanding without help in the proposed tasks 4 Run pilot and eight sessions Complete records, with documented limitations 5 Iterate and check again No critical crashes; updated WCAG matrix References [1] W3C. Web Content Accessibility Guidelines (WCAG) 2.2.

https://www.w3.org/TR/WCAG22/ [2] W3C. Understanding Success Criterion 1.4.3: Contrast (Minimum).

Official Minimum Contrast Guide [3] European Union. Regulation (EU) 2016/679. Articles 5, 6, 9, 13 and 32.

Official text of the GDPR in EUR-Lex Evaluated material: Cala HTML file contributed by the user. References consulted on October 3 of 2026.

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A. Observation sheet field Content to be recorded Participant/task Pseudonymized code and task number Settings Device, browser and supports used Result Without help / with help / abandonment / not evaluable Time and help Duration when applicable and specific help Evidence Observed action or deidentified comment Barrier and severity Problem, consequence and priority Ease Rating from 1 to 7

B. Subsequent questions

1. What has been most difficult or confusing for you?

2. What amount do you think you would pay and what does it include?

3. What information would you need before confirming?

4. Have you encountered any barriers with your usual way of using the device?

5. From 1 to 7, how easy was this task?

C. Prior information and consent Before participating, an accessible sheet will be delivered with the person responsible and contact, purpose, legal basis, recipients, deadlines and rights. It will be explained that there are no real payments, that participating is voluntary and that you can stop the session. Recording authorization will be collected separately. This annex is a script of preparation; must be specified before collecting data.

D. Accessibility matrix For each applicable criterion, the following will be recorded: WCAG identifier, screen, method, evidence, result. (complies / does not comply / does not apply / pending), action and subsequent verification. AA will not be declared while there are non-compliances or pending applicable verifications.